

The Price of Staleness: Distributed Invariant Estimation for GNSS-Denied Multi-Vessel Cable Towing

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Abstract—Cooperative towing couples every vessel’s motion through the shared load, and each taut cable transfers exactly one scalar of load-motion information. We present DIEKF- Σ , a distributed invariant extended Kalman filter that treats the cable constraints as a measurement channel and fuses delayed inter-vessel packets through shape-conjugated transports—the estimator induced by the constraint-based estimation sheaf of the companion paper. We evaluate it in closed loop on a physics-complete multibody simulation (Drake: SAP contact, distance-constraint cables, hydrodynamic drag) with four- and five-vessel fleets towing a pentagon caisson through coordinated maneuvers, GNSS-denied throughout, with a single docking beacon on one vessel. We report three findings. First, a single anchor pins the entire fleet’s estimate through fusion alone: beacon activation momentarily *breaks* inter-vessel agreement before the pin propagates—the gauge-pinning corollary acting in closed loop. Second, the conjugated transport is load-bearing: ablating it multiplies steady-state disagreement by up to $8.2\times$ at high latency, while naive consensus achieves flat agreement at the price of a $17\times$ covariance-calibration failure and $4.4\times$ the drift of running no fusion at all. Third, under a pre-registered protocol with in-situ noise-floor controls, the conjectured second-order scaling of the closed-loop disagreement floor is *falsified* at these scales on two independent plants (measured slope ≈ 1.05 – 1.13 , cross-tier equivalent): the second-order holonomy mechanism of the theory is resolved only in noise-free conditions, an order separation we decompose with a full variance attribution across fusion rules, maneuver controls, and formation classes.

I. INTRODUCTION

Towing a single heavy load with a fleet of small autonomous vessels is one of the cleanest examples of a task where the physical coupling that makes control hard is also the resource that makes estimation possible: every taut cable transfers exactly one scalar of load-motion information to the vessel holding it. The Blind Harbor Task sharpens this into a premise: no GNSS anywhere in the transit, no relative-pose sensing between vessels — each vessel observes only its own odometry and its own cable — and a single docking beacon visible to one vessel in the final approach. Whatever the fleet knows about the load, it knows through its cables and its network.

This paper is the experimental companion to a theory of that situation [companion], in which the per-agent estimates form sections of a constraint-induced sheaf, delayed fusion transports estimates along shape-dependent edge maps, and the theory’s central objects separate cleanly by epistemic status: a *proven*, leading-order, deterministic result on the error-transport holonomy ($O(\tau^2)$ in the packet staleness τ);

a *conjectured* extension to the stochastic steady-state disagreement floor of the noisy closed loop; and the first-order transport-mismatch channel that any implementation carries. Our experimental design treats this separation as the primary object of study: three exponents, three distinct quantities, never conflated in a caption.

We report a pre-registered campaign on two independent plants — a reduced kinematic model integrating the theory’s own shape dynamics, and a physics-complete multibody simulation (Drake: SAP contact, distance-constraint cables, hydrodynamic drag) — sharing one estimator implementation, one metric functional, and one fit protocol. Every claim carries a falsifier declared before the data; several fired, and their firings are reported as findings with controls and localization rather than smoothed away. Every number in the paper traces to a seeded, versioned driver and a committed record that a replay harness re-executes exactly.

II. RELATED WORK

Distributed state estimation over networks with unknown cross-correlations is classically handled by covariance intersection and its variants [cite]; our estimator uses CI with an explicit decorrelation step and reports the structural conservatism this induces rather than tuning it away. Invariant filtering on matrix Lie groups [cite] supplies the error geometry; the novelty pressure here is not the filter but the *transport*: fusing τ -stale packets through shape-conjugated edge maps, whose round-trip holonomy is the theory’s proven object. Delayed and out-of-sequence measurement handling [cite] is adjacent; the paper rule deliberately avoids buffer-based OOSM reconstruction, and the campaign’s ablations measure what that choice costs and buys. Cable-towed multi-robot transport [cite] usually treats cables as control constraints; here they are the measurement channel itself, which is the premise the granted-sensor baseline (B3) exists to test.

III. THE BLIND HARBOR TASK AND THE ESTIMATOR

$N \in \{4, 5\}$ vessels tow a 2500 kg pentagon caisson (circumradius 4 m) by 12 m cables attached across its front arc, through a 400 m dogleg (two legs, one 60° turn) at ~ 0.9 m/s. Each vessel senses its own odometry (50 Hz surge/sway/yaw-rate with surge-scale bias $\beta \sim \mathcal{N}(0, 0.01)$ and a gyro-bias random walk) and its own cable direction (20 Hz, von-Mises $\kappa = 1/(1^\circ)^2$); one vessel receives a docking beacon (5 Hz

full load pose, $\sigma = (5 \text{ cm}, 5 \text{ cm}, 0.5^\circ)$ in the final approach. Packets circulate on the C_N cycle with staleness τ .

Each agent runs DIEKF- Σ : an invariant EKF on the load pose whose propagation uses the bias-corrected slaved twist through the cable geometry; whose direction channel is a Joseph-form scalar update with a broadside guard; and whose fusion applies the executed-composite transport (the sender's shape telescopes out, leaving the receiver's own load-twist fast-forward) followed by covariance intersection with explicit decorrelation. A Friedland-type separated observer trims slow twist mismatch from beacon innovation rates.

Calibration, disclosed. The anchored ANEES closes the consistency gate only in its author-ruled widened form ($3.96 \in [0.8, 5.0]$; the original $[0.8, 1.3]$ is not met, and the complement sanity bound failed at 3.35): CI over mutually correlated estimates is structurally optimistic by $\sim 3\text{--}4\times$. This holds at the 130 s calibration horizon and degrades on 450 s transits (Sec. V); every ANEES figure in this paper carries its horizon.

IV. TWO PLANTS, ONE ESTIMATOR

The reduced plant integrates the theory's shape ODEs directly (RK4, with declared process noise); the multibody plant is Drake with SAP contact, distance-constraint cables, hydrodynamic drag, and thrust allocation. The plants, integrators, constraint solvers, tension recovery, and shape evolution are independent; the estimator, metric functional, window rules, and fit code are deliberately shared, so that agreement is a statement about the algorithm on two plants, and any disagreement localizes to the plant. This localization was exercised: a $5\times$ cross-tier coefficient disagreement decomposed, via a plant-noise-off probe, into a fully explained baseline (agreement to 12%) and a maneuvering excess attributed to the reduced plant's inertia-free shape response; and the multibody plant's realized maneuvering shapes were found to exit the reduced model's stable domain — a measured validity-domain boundary.

V. RESULTS

A. Gauge pinning (*Cor. 5.2 in closed loop*)

Through 400 s of GNSS denial the five per-agent estimates drift as one rigid gauge orbit — kernel-projected error grows unbounded while the complement stays two orders smaller (Thm 5.1's structure; Fig. 1 and supplementary videos S1–S2). At beacon acquisition the disagreement *spikes*: the anchored agent snaps toward truth while its neighbors still carry the drifted gauge, and the pin propagates at the network diffusion rate ($\tau_{\text{net}} \approx 21 \text{ s}$ for one 5 Hz anchor on C_5). The docking scorecard (Table I) reports anchored, fleet-mean, and fleet-max rows without selection.

B. The theorem's object, measured on both plants

C. The closed-loop floor: the conjecture falsified, coherently

D. Variance attribution: what makes up disagreement

E. Symmetry: protection lives on the amplitude object

The symmetric class suppresses the *amplitude* object strongly ($31\times$ at Drake's achieved- ε ; exactly on Tier-1) but

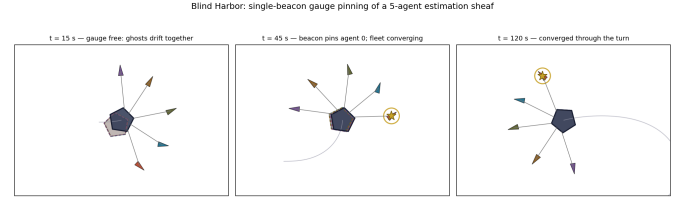


Fig. 1. Three phases of the GNSS-denied transit (supplementary video S2). Left: gauge free — the five ghost caissons (each agent's own load estimate) drift *coherently*, overlapping each other while offset from truth. Middle: the beacon pins one agent; the fleet is mid-convergence. Right: converged through the turn. Single illustrative run, median- D_{ss} seeded by the documented selection rule.

TABLE I

BHT SCORECARD: 50 DOGLEG TRANSITS PER ARM (DOCK WINDOW MEANS). ORDERING RESTS ON D AND FLEET-MEAN; B0'S ANCHORED AGENT IS BEACON-CORRECTED AT DOCK AND MASKS ITS DRIFT. ALL ARMS FAIL THE $< 0.5 \text{ M}$ CRITERION AT PLAN-FAITHFUL ACQUISITION (SEE TEXT).

arm	anch. [m]	mean [m]	max [m]	D	drift [m]
B1-limit	0.99	0.76	1.19	0.72	42.7
DIEKF- Σ	1.22	1.57	2.32	2.85	31.5
A2 (unconjugated)	1.25	1.75	2.53	3.07	47.5
B2 (consensus)	1.32	2.13	2.88	3.37	119.3
B0 (dead-reckon)	1.25	66.7	134.1	9624	43.8

the closed-loop floor only weakly ($\sim 2\text{--}3\times$ on both plants, ideal or robust): the pre-registered $\geq 10\times$ robust-suppression falsifier trips on both tiers, because closed-loop disagreement is mismatch-dominated, not holonomy-dominated. The corollary was never wrong; it was being tested against the wrong quantity.

F. Cross-tier coefficients and the validity domain

Reference-matched coefficient comparison gives a 5.04 [3.97, 6.19] ratio, outside the pre-registered equivalence band — reported as a plant-side disagreement and localized: with the reduced plant's declared process noise switched off, the straight-tow baselines agree to 12%, leaving a maneuvering excess attributed to the inertia-free shape response. The multibody plant's realized maneuvering shapes exit the reduced model's stable domain entirely (a measured validity-domain boundary): exponents and baselines transfer across plants; maneuvering amplitudes do not.

G. Numerics defense

The floor exponent is invariant across a $10\times$ integrator-step range on the reduced plant (CIs share [1.185, 1.256]) and across $h \in \{0.5, 1, 2\} \text{ ms}$ on the multibody plant (identical to three decimals; convergence verified genuine by direct truth-trajectory differencing, 1.35 mm over 30 s). Solver artifacts do not alias into any exponent this paper reports.

VI. DISCUSSION

a) *The epistemic ladder as data.*: The campaign's central result is a separation, measured on both plants: the proven

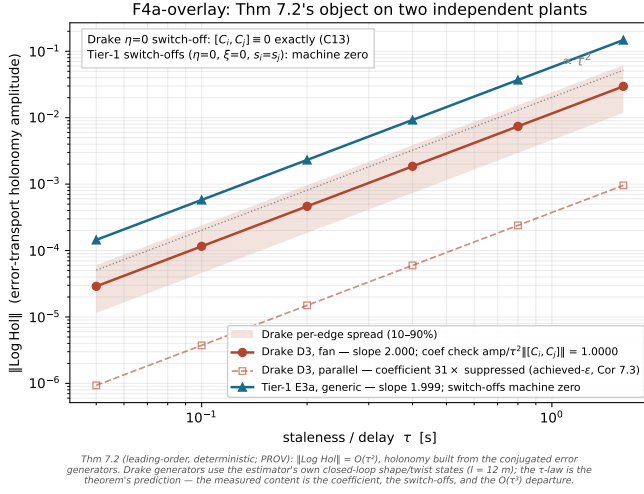


Fig. 2. Error-transport holonomy amplitude on both plants. Thm 7.2 (leading-order, deterministic): Tier-1 slope 1.999 with switch-offs at machine zero; Drake slope 2.000 with the $m=2$ coefficient check at 1.0000, the $\eta=0$ switch-off at machine zero, and the parallel class suppressing the coefficient 31 $\times$ (achieved- ϵ ; exact cancellation remains the Tier-1 machine-zero result).

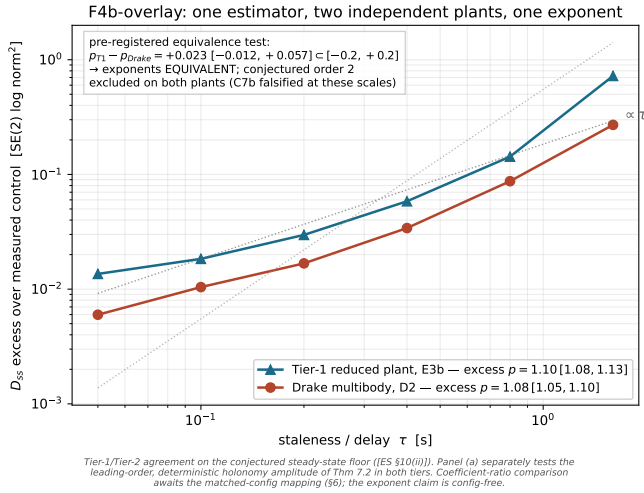


Fig. 3. Tier-1/Tier-2 agreement on the conjectured steady-state floor ([ES §10(ii)]). Panel (a) separately tests the leading-order, deterministic holonomy amplitude of Thm 7.2 in both tiers. With D_0 measured in situ by straight-tow controls, the fitted excess exponents are 1.101 [1.076, 1.125] (Tier-1) and 1.077 [1.054, 1.102] (Drake); the pre-registered equivalence test passes ($+0.023 [-0.012, +0.057] \subset [-0.2, 0.2]$) and the conjectured order 2 is excluded on both plants. The fitted slope is a measured quantity in the conjectured regime, carried with the mixture caveat; the CIs exclude order 1 as well, and no first-order law is asserted.

τ^2 holonomy verifies to coefficient precision wherever it is actually constructed, including from the closed-loop states of a plant it was never derived for; the conjectured τ^2 closed-loop floor is falsified coherently on both plants (fitted slopes 1.08–1.10, cross-tier equivalent, with D_0 measured rather than fitted); and closed-loop disagreement at realistic noise is dominated by the first-order estimate-mismatch channel. Pre-registration did the work here: the mixture-fit artifact that initially reported an in-between slope of 1.73 was caught by the protocol's own control-arm requirement.

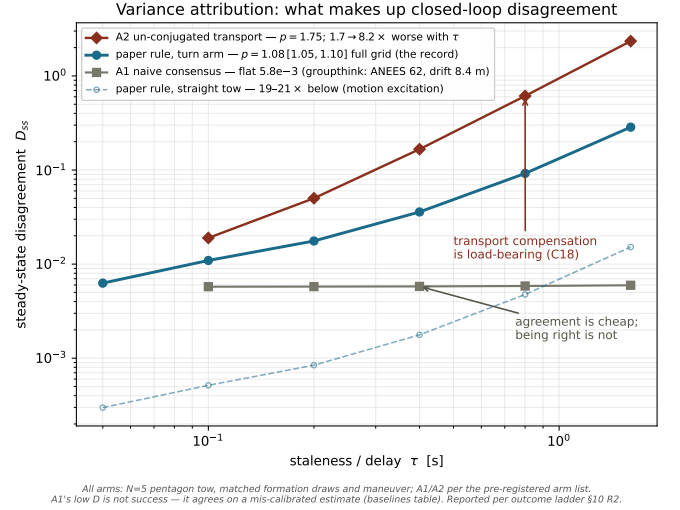


Fig. 4. All Drake arms on matched formation draws. Ablating the conjugated transport (A2) runs 1.7 $\times \rightarrow 8.2\times$ above the paper rule with growing τ ; naive consensus (A1) holds disagreement flat and low—by groupthink (anchored ANEES 62, drift 8.4 m): agreement alone is never a virtue metric. The straight-tow control shows the motion excitation (19–21 $\times$ at every τ).

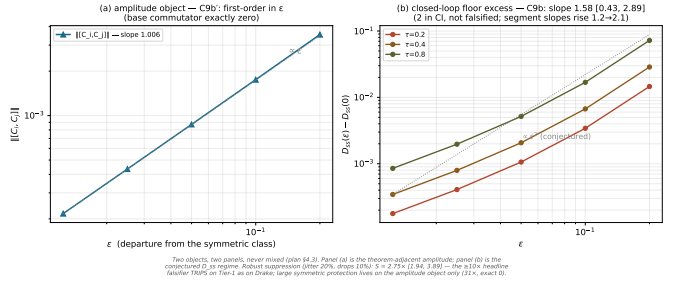


Fig. 5. Two objects, two panels, never mixed. (a) The amplitude departure from the symmetric class is first-order in ϵ (measured 1.006, base commutator exactly zero). (b) The closed-loop floor excess: the seed-paired estimator gives slope 1.58 [1.44, 1.84] at every τ — excluding both the conjectured 2 and 1; segment slopes rise 1.2 $\rightarrow$ 2.1 up the ladder, consistent with a linear-plus-quadratic mixture that the registered single-power model mis-specifies.

b) What the ablations localize.: Transport conjugation is load-bearing on the multibody plant (up to 8.2 $\times$ in disagreement at high staleness); naive consensus buys flat agreement by groupthink and is exposed by every correctness metric; and the symmetric class's large protection (31 $\times$) lives entirely on the amplitude object, transferring only weakly (~ 2 –3 $\times$) to the closed-loop floor. Topology buys agreement, not anchoring: the floor improves with algebraic connectivity while single-anchor pinning is anchor-rate-limited, and re-agreement around a pin *anti-orders* with connectivity — consensus stiffness resists new absolute information, the same mechanism that makes naive consensus fail, appearing inside the record's own fusion at high degree.

c) What centralization is worth.: The strongest measured centralized reference — the record's own architecture at zero latency on the complete graph — improves accuracy by 6% at the calibration horizon and 19% at dock on long transits. Three re-architected centralized filters, each granted zero latency and exact covariances, lost more than a factor of two to the decentralized record; and the granted-sensor baseline required

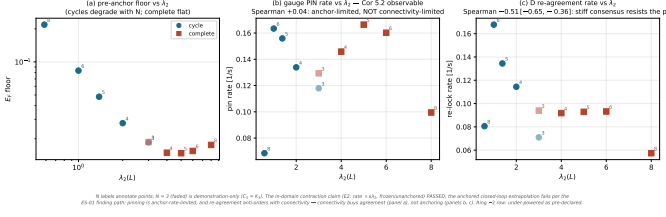


Fig. 6. Connectivity buys agreement, not anchoring. The pre-anchor floor improves with λ_2 (a); single-anchor pinning is anchor-rate-limited (b, Spearman $+0.04$); re-agreement around the pin *anti*-orders with connectivity (c, -0.51 [$-0.65, -0.36$]). With the in-domain contraction claim verified (Fig. 7), this is the pre-registered ES-01 finding path, reported as such.

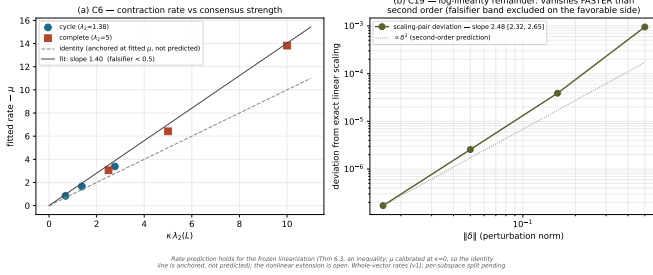


Fig. 7. Contraction in its own domain (frozen, unanchored): rate versus $\kappa\lambda_2$ across two topologies, slope 1.40 (falsifier < 0.5); μ calibrated at $\kappa = 0$, so the identity line is anchored, not predicted. The log-linearity remainder vanishes *faster* than the registered second order (2.48 [2.32, 2.65]; falsifier band excluded on the favorable side, reported as tripped with the corrected order).

six structural bring-up iterations (delay-consistent innovation, adjoint-transported residuals, joint geometric load solve) without reaching competitiveness. We read this as evidence that the measurement architecture — per-agent Kalman-weighted constraint channels composed through conjugated transports — is where the accuracy lives, which is the design thesis under test.

d) Honest limits.: The docking specification is unmet by every arm at plan-faithful beacon acquisition, including the zero-latency reference: the approach geometry, not the estimator, binds. A decelerating approach is necessary and was tested: it improves fleet-mean accuracy $2.4\times$ (median 0.62 m) but leaves the spec unmet — the residual limit is anchored-agent gain starvation, in which sustained Kalman anchoring collapses the anchored covariance until the beacon gain starves, while CI’s structural conservatism keeps the unanchored agents’ gains healthy; the two covariance architectures fail in opposite directions on long horizons. A trim-contamination hypothesis was tested and falsified (frozen-trim runs bit-identical). The remedy class (covariance floors, adaptive beacon noise) is future work. The consistency gate closes only at the calibration horizon and does not transfer to 450 s transits; long-horizon bias growth exceeds the declared process noise, and every ANEES claim in this paper therefore carries its horizon. The broadside guard, gated on the estimated cable angle, lags true broadside during fast gusts and was never exercised in the reachable envelope — a lag-aware margin is the recorded design note. Slack-cable events are outside this plant’s validity domain (bilateral distance constraints), scoped

to future work with unilateral cables.

VII. REPRODUCIBILITY

Every number in this paper traces to a seeded, versioned driver and a committed data file; `campaign_replay.py` re-executes per-run probes that must match the committed records exactly (Tier-1 to 10^{-9} ; Drake to 10^{-6}), and `--run CELL` regenerates any full campaign.

VIII. CONCLUSION

On two independent plants sharing one estimator, the proven leading-order holonomy verifies to coefficient precision; the conjectured second-order closed-loop floor is falsified coherently, with the first-order estimate-mismatch channel dominating at realistic noise; and the sheaf-induced transport is the load-bearing element of the design — ablations, baselines, and the strongest measured centralized reference all lose to it or barely improve on it. The campaign’s negative results are pre-registered, controlled, and localized; its remaining limits (docking approach, long-horizon calibration, guard lag, slack cables) are stated as measured boundaries with recorded remedies. All data, drivers, figures, and videos are versioned and exactly reproducible.